

2nd Sessional Test- Heat Transfer 06.04.2026

1. An object of irregular shape has a characteristic length of $L = 1.0$ m and is maintained at a uniform surface temperature of $T_s = 400$ K. When placed in atmospheric air at a temperature of $T_\infty = 300$ K and moving with a velocity of $V = 100$ m/s, the average heat flux from the surface to the air is $20,000$ W/m². If a second object of the same shape, but with a characteristic length of $L = 5$ m, is maintained at a surface temperature of $T_s = 400$ K and is placed in atmospheric air at $T_\infty = 300$ K, what will the value of the average convection coefficient be if the air velocity is $V = 20$ m/s? (8)
2. Fully developed conditions are known to exist for water flowing through a 25-mm-diameter tube at 0.01 kg/s and 27°C . What is the maximum velocity of the water in the tube? Water (300K): $\rho = 998$ kg/m³, $\mu = 855 \times 10^{-6}$ N·s/m² (7)